

261. If the coefficients of x^2 and x^3 are both zero, in the expansion of the expression $(1 + ax + bx^2)(1 - 3x)^{15}$ in powers of x , then the ordered pair (a, b) is equal to :

- (A) $(-54, 315)$ (B) $(28, 861)$ (C) $(-21, 714)$ (D) $(28, 315)$

यदि x की घातों (powers) में, व्यंजक $(1 + ax + bx^2)(1 - 3x)^{15}$ के प्रसार में x^2 तथा x^3 दोनों के गुणांक शून्य के बराबर हैं, तो क्रमित युग्म (a, b) बराबर है :

- (A) $(-54, 315)$ (B) $(28, 861)$ (C) $(-21, 714)$ (D) $(28, 315)$

Ans. D

Sol. $(1 + ax + bx^2)(1 - 3x)^{15}$

$$\text{Co-eff. of } x^2 = 1 \cdot {}^{15}C_2(-3)^2 + a \cdot {}^{15}C_1(-3) + b \cdot {}^{15}C_0$$

$$= \frac{15 \times 14}{2} \times 9 - 15 \times 3a + b = 0 \text{ (Given)}$$

$$\Rightarrow 945 - 45a + b = 0 \quad \dots(i)$$

Now co-eff. of $x^3 = 0$

$$\Rightarrow {}^{15}C_3(-3)^3 + a \cdot {}^{15}C_2(-3)^2 + b \cdot {}^{15}C_1(-3) = 0$$

$$\Rightarrow \frac{15 \times 14 \times 13}{3 \times 2} \times (-3 \times 3 \times 3) + a \times \frac{15 \times 14 \times 9}{2} - b \times 3 \times 15 = 0$$

$$\Rightarrow 15 \times 3[-3 \times 7 \times 13 + a \times 7 \times 3 - b] = 0$$

$$\Rightarrow 21a - b = 273 \quad \dots(ii)$$

From (i) and (ii)

$$a = +28, b = 315 \equiv (a, b) \equiv (28, 315)$$

262. The smallest natural number n , such that the coefficient of x in the expansion of $\left(x^2 + \frac{1}{x^3}\right)^n$ is ${}^nC_{23}$, is :

- (A) 58 (B) 35 (C) 38 (D) 23

वह न्यूनतम प्राकृत संख्या n , जिसके लिए $\left(x^2 + \frac{1}{x^3}\right)^n$ के प्रसार में x का गुणांक ${}^nC_{23}$ है, है :

- (A) 58 (B) 35 (C) 38 (D) 23

Ans. C

Sol. $\left(x^2 + \frac{1}{x^3}\right)^n$

$$\text{General term } T_{r+1} = {}^nC_r (x^2)^{n-r} \left(\frac{1}{x^3}\right)^r$$

$${}^nC_r \cdot x^{2n-5r}$$

for coefficient of x , $2n - 5r = 1$

Given ${}^nC_r = {}^nC_{23}$

$$r = 23 \quad \text{or} \quad n - r = 23$$

$$\Rightarrow n = 58 \quad \text{or} \quad n = 38$$

Minimum value is $n = 38$

263. The coefficient of x^{18} in the product $(1+x)(1-x)^{10}(1+x+x^2)^9$ is :

- (A) 84 (B) -126 (C) -84 (D) 126

गुणनफल $(1+x)(1-x)^{10}(1+x+x^2)^9$ में x^{18} का गुणांक है :

- (A) 84 (B) -126 (C) -84 (D) 126

Ans. A

Sol. $(1-x)^{10}(1+x+x^2)^9(1 \times x)$

$$= (1-x^3)^9(1-x^2)$$

$$= (1-x^3)^9 - x^2(1-x^3)^9$$

$\Rightarrow$ Coefficient of x^{18} in $(1-x^3)^9$ - coeff. of x^{16} in

$$(1-x^3)^9$$

$${}^9C_6 = \frac{9!}{6!3!} = \frac{7 \times 8 \times 9}{6} = 84$$

264. If ${}^{20}C_1 + (2^2) {}^{20}C_2 + (3^2) {}^{20}C_3 + \dots + (20^2) {}^{20}C_{20} = A(2^\beta)$, then the ordered pair (A, β) is equal to

- (A) (420,19) (B) (380,19) (C) (420,18) (D) (380,18)

यदि ${}^{20}C_1 + (2^2) {}^{20}C_2 + (3^2) {}^{20}C_3 + \dots + (20^2) {}^{20}C_{20} = A(2^\beta)$, तो क्रमित युग्म (A, β) बराबर है :

- (A) (420,19) (B) (380,19) (C) (420,18) (D) (380,18)

Ans. C

Sol. ${}^{20}C_1 + 2^2 \cdot {}^{20}C_2 + 3^2 \cdot {}^{20}C_3 + \dots + 20^2 \cdot {}^{20}C_{20}$

$$= \sum_{r=1}^{20} r^2 \cdot {}^{20}C_r$$

$$= 20 \sum_{r=1}^{20} r \cdot {}^{19}C_{r-1}$$

$$= 20 \left[\sum_{r=1}^{20} (r-1) {}^{19}C_{r-1} + \sum_{r=1}^{20} {}^{19}C_{r-1} \right]$$

$$= 20 \left[19 \sum_{r=2}^{20} {}^{18}C_{r-2} + 2^{19} \right]$$

$$= 20[19 \cdot 2^{18} + 2^{19}]$$

$$= 20 \times 21 \times 2^{18}$$

$$= 420 \times 2^{18}$$

So; $A = 420$ and $\beta = 18$

265. The term independent of x in expansion of $\left(\frac{1}{60} - \frac{x^8}{81}\right) \cdot \left(2x^2 - \frac{3}{x^2}\right)^6$ is equal to

- (A) -108 (B) -36 (C) -72 (D) 36

$\left(\frac{1}{60} - \frac{x^8}{81}\right) \cdot \left(2x^2 - \frac{3}{x^2}\right)^6$ के प्रसार में x से स्वतंत्र पद है :

- (A) -108 (B) -36 (C) -72 (D) 36

Ans. B

Sol. $\left(\frac{1}{60} - \frac{x^8}{81}\right) \cdot \left(2x^2 - \frac{3}{x^2}\right)^6$

$$= \frac{1}{60} \left(2x^2 - \frac{3}{x^2}\right)^6 - \frac{x^8}{81} \left(2x^2 - \frac{3}{x^2}\right)^6$$

$$\text{Coefficient of } x^0 \text{ in } = \frac{1}{60} \left(2x^2 - \frac{3}{x^2}\right)^6 - \frac{x^8}{81}$$

$$\text{coefficient of } x^{-8} \text{ in } \left(2x^2 - \frac{3}{x^2}\right)^6$$

$$= \frac{-1}{60} {}^6C_3 (2)^3 (3)^3 + \frac{1}{81} {}^6C_5 (2)(3)^5$$

$$= -72 + 36$$

$$= -36$$

266. The greatest positive integer k , for which $49^k + 1$ is a factor of the sum $49^{125} + 49^{124} + \dots + 49^2 + 49 + 1$, is :

- (A) 60 (B) 63 (C) 65 (D) 32

सबसे बड़ी धन पूर्णांक संख्या k , जिसके लिए $49^k + 1$ योगफल $49^{125} + 49^{124} + \dots + 49^2 + 49 + 1$ का एक गुणन खंड है, है :

- (A) 60 (B) 63 (C) 65 (D) 32

Ans. B

Sol. $\text{Sum} = \frac{49^{126} - 1}{49 - 1} = \frac{(49^{63} + 1)(49^{63} - 1)}{49 - 1}$

$$\text{So } k = 63$$

267. The number of ordered pairs (r, k) for which $6 \cdot {}^{35}C_r = (k^2 - 3) \cdot {}^{36}C_{r+1}$, where k is an integer, is :

- (A) 3 (B) 4 (C) 2 (D) 6

क्रमित युग्मों (r, k) , जिनके लिए $6 \cdot {}^{35}C_r = (k^2 - 3) \cdot {}^{36}C_{r+1}$, जहाँ k एक पूर्णांक है, की संख्या है :

- (A) 3 (B) 4 (C) 2 (D) 6

Ans. B

Sol. $(6) \binom{35}{r} = k^2 - 3 \binom{36}{r+1}$

$$\frac{k^2 - 3}{6} = \frac{\binom{35}{r}}{\binom{36}{r+1}}$$

$$\frac{k^2 - 3}{6} = \frac{r+1}{36}$$

$$6k^2 = r + 19$$

$$k = 2 \Rightarrow r = \pm 5$$

$$k = -2 \Rightarrow r = 5$$

$$k = 3 \Rightarrow r = 35$$

$$k = -3 \Rightarrow r = 35$$

Total no. of ordered pairs (r, k) is 4

268. If α and β be the coefficients of x^4 and x^2 respectively in the expansion of $(x + \sqrt{x^2 - 1})^6 + (x - \sqrt{x^2 - 1})^6$, then:

- (A) $\alpha + \beta = -30$ (B) $\alpha + \beta = 60$ (C) $\alpha - \beta = 60$ (D) $\alpha - \beta = -132$

यदि $(x + \sqrt{x^2 - 1})^6 + (x - \sqrt{x^2 - 1})^6$ के प्रसार में x^4 तथा x^2 के गुणांक क्रमशः α तथा β हैं, तो :

- (A) $\alpha + \beta = -30$ (B) $\alpha + \beta = 60$ (C) $\alpha - \beta = 60$ (D) $\alpha - \beta = -132$

Ans. D

Sol. $(x + \sqrt{x^2 - 1})^6 + (x - \sqrt{x^2 - 1})^6 = 2 \left[{}^6C_0 x^6 + {}^6C_2 x^4 (x^2 - 1) + {}^6C_4 x^2 (x^2 - 1) + {}^6C_6 (x^2 - 1)^3 \right]$

$$= 2 \left[32x^6 - 48x^4 + 18x^2 - 1 \right]$$

Coefficient of x^2 (α) = -96

Coefficient of x^2 (β) = 36

$$\alpha - \beta = -132$$

269. In the expansion of $\left(\frac{x}{\cos \theta} + \frac{1}{x \sin \theta} \right)^{16}$, if l_1 is the least value of the term independent of x when $\frac{\pi}{8} \leq \theta \leq \frac{\pi}{4}$

and l_2 is the least value of the term independent of x when $\frac{\pi}{16} \leq \theta \leq \frac{\pi}{8}$, then the ratio $l_2 : l_1$ is equal to :

- (A) 1 : 16 (B) 16 : 1 (C) 1 : 8 (D) 8 : 1

$\left(\frac{x}{\cos \theta} + \frac{1}{x \sin \theta} \right)^{16}$ के प्रसार में, यदि x से स्वतंत्र पद का निम्नतम मान l_1 है जब $\frac{\pi}{8} \leq \theta \leq \frac{\pi}{4}$ तथा x से स्वतंत्र पद का निम्नतम

मान l_2 है जब $\frac{\pi}{16} \leq \theta \leq \frac{\pi}{8}$, तो अनुपात $l_2 : l_1$ बराबर है :

- (A) 1 : 16 (B) 16 : 1 (C) 1 : 8 (D) 8 : 1

Ans. B

Sol. $T_{r+1} = {}^{16}C_r \left(\frac{x}{\cos \theta} \right)^{16-r} \left(\frac{1}{x \sin \theta} \right)^r$
 $= {}^{16}C_r x^{16-2r} (\cos \theta)^{r-16} (\sin \theta)^{-r}$
 $= 16 - 2r = 0 \Rightarrow r = 8$
 $= {}^{16}C_8 (\cos \theta)^{-8} (\sin \theta)^{-8}$

Term independent of $x = {}^{16}C_8 \times \frac{1}{(\sin \theta \cos \theta)^8}$
 $= {}^{16}C_8 \times \frac{2^8}{(\sin 2\theta)^8}$

$l_1 : \frac{\pi}{4} \leq 2\theta \leq \frac{\pi}{2} \quad l_2 : \frac{\pi}{8} \leq 2\theta \leq \frac{\pi}{4}$
 $l_1 : {}^{16}C_8 \times 2^8 \quad l_2 : {}^{16}C_8 \times 2^{12}$
 $l_2 : l_1 \Rightarrow 2^4 : 1 \Rightarrow 16 : 1$

270. The value of

$-{}^{15}C_1 + 2 \cdot {}^{15}C_2 - 3 \cdot {}^{15}C_3 + \dots - 15 \cdot {}^{15}C_{15} + {}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 + \dots + {}^{14}C_{11}$ is :

$-{}^{15}C_1 + 2 \cdot {}^{15}C_2 - 3 \cdot {}^{15}C_3 + \dots - 15 \cdot {}^{15}C_{15} + {}^{14}C_1 + {}^{14}C_3 + {}^{14}C_5 + \dots + {}^{14}C_{11}$ का मान है :

(A) $2^{13} - 14$ (B) 2^{14} (C) $2^{16} - 1$ (D) $2^{13} - 13$

Ans. A

Sol. $(1+x)^{15} = {}^{15}C_0 + {}^{15}C_1x + {}^{15}C_2x^2 + \dots + {}^{15}C_{15}x^{15}$

D w r to x

$15(1+x)^{14} = {}^{15}C_1 + 2 \cdot {}^{15}C_2x + \dots + 15 \cdot {}^{15}C_{15}x^{14}$

Put $x = -1$

$-{}^{15}C_1 + 2 \cdot {}^{15}C_2 + \dots - 15 \cdot {}^{15}C_{15} = 0$

${}^{14}C_1 + {}^{14}C_3 + \dots + {}^{14}C_{11} + {}^{14}C_{13} = 2^{13}$

${}^{14}C_1 + {}^{14}C_3 + \dots + {}^{14}C_{11} = 2^{13} - 14$

$S = 2^{13} - 14$

271. If $n \geq 2$ is a positive integer, then the sum of the series ${}^{n+1}C_2 + 2({}^2C_2 + {}^3C_2 + {}^4C_2 + \dots + {}^nC_2)$ is :

यदि $n \geq 2$ एक धनात्मक पूर्णांक है, तो श्रेणी ${}^{n+1}C_2 + 2({}^2C_2 + {}^3C_2 + {}^4C_2 + \dots + {}^nC_2)$ का योग है :

(A) $\frac{n(n+1)^2(n+2)}{12}$ (B) $\frac{n(n+1)(2n+1)}{6}$

(C) $\frac{n(2n+1)(3n+1)}{6}$

(D) $\frac{n(n-1)(2n+1)}{6}$

Ans. B

Sol. $E = {}^{n+1}C_2 + 2({}^3C_3 + {}^3C_2 + {}^4C_2 + \dots + {}^nC_2)$
 $= {}^{n+1}C_2 + 2({}^{n+1}C_3)$
 $= \frac{(n+1)n}{2} + \frac{2(n+1)n(n-1)}{6}$
 $= \frac{n(n+1)(3+2n-2)}{6}$
 $= \frac{n(n+1)(2n+1)}{6} \Rightarrow \text{option (2)}$

272. The maximum value of the term independent of 't' in the expansion of $\left(tx^{\frac{1}{5}} + \frac{(1-x)^{\frac{1}{10}}}{t} \right)^{10}$

where $x \in (0, 1)$ is :

$\left(tx^{\frac{1}{5}} + \frac{(1-x)^{\frac{1}{10}}}{t} \right)^{10}$, जहाँ $x \in (0, 1)$ है, के प्रसार में 't' से स्वतंत्र पद का अधिकतम मान है :

(A) $\frac{2 \cdot 10!}{3\sqrt{3}(5!)^2}$

(B) $\frac{2 \cdot 10!}{3(5!)^2}$

(C) $\frac{10!}{\sqrt{3}(5!)^2}$

(D) $\frac{10!}{\sqrt{3}(5!)^2}$

Ans. A

Sol. The term independent of t will be middle term :

Middle term $f(x) = {}^{10}C_5 \cdot x \cdot (1-x)^{\frac{1}{2}} ; x \in (0, 1)$.

$f'(x) = {}^{10}C_5 \cdot \left[(1-x)^{\frac{1}{2}} - \frac{x}{2}(1-x)^{-\frac{1}{2}} \right] = 0$

$\Rightarrow 1-x = \frac{x}{2} \Rightarrow \frac{3x}{2} = 1 \Rightarrow x = \frac{2}{3}$.

Point of maxima.

$f\left(\frac{2}{3}\right) = {}^{10}C_5 \times \frac{2}{3} \left(1 - \frac{2}{3}\right)^{\frac{1}{2}}$

$$= \frac{2}{3\sqrt{3}} \times \frac{10!}{5!5!}$$

273. Let $[x]$ denote greatest integer less than or equal to x . If for $n \in \mathbb{N}$, then $(1 - x + x^3)^n = \sum_{j=0}^{3n} a_j x^j$, then

$$\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j} + 4 \sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j+1} \text{ is equal to :}$$

माना $[x]$ महत्तम पूर्णांक $\leq x$ है। यदि $n \in \mathbb{N}$ के लिए $(1 - x + x^3)^n = \sum_{j=0}^{3n} a_j x^j$ है, तो $\sum_{j=0}^{\left[\frac{3n}{2}\right]} a_{2j} + 4 \sum_{j=0}^{\left[\frac{3n-1}{2}\right]} a_{2j+1}$ बराबर है:

- (A) n (B) 2^{n-1} (C) 2 (D) 1

Ans. D

Sol. $f(x) = (1 - x + x^3)^n$

$$f(1) = 1, f(-1) = 1$$

$$\sum a_{2j} = \frac{f(1) + f(-1)}{2} = \frac{1+1}{2} = 1$$

$$\sum a_{2j+1} = \frac{f(1) - f(-1)}{2} = \frac{1-1}{2} = 0$$

$$\text{Required sum} = 1 + 4(0) = 1$$

274. If $({}^{30}C_1)^2 + 2({}^{30}C_2)^2 + 3({}^{30}C_3)^2 + \dots + 30({}^{30}C_{30})^2 = \frac{\alpha 60!}{(30!)^2}$ then α is equal to :

यदि $({}^{30}C_1)^2 + 2({}^{30}C_2)^2 + 3({}^{30}C_3)^2 + \dots + 30({}^{30}C_{30})^2 = \frac{\alpha 60!}{(30!)^2}$ है, तो α बराबर है :

- (A) 15 (B) 60 (C) 10 (D) 30

Ans. A

Sol. $S = 0 \cdot ({}^{30}C_0)^2 + 1 \cdot ({}^{30}C_1)^2 + 2 \cdot ({}^{30}C_2)^2 + \dots + 30 \cdot ({}^{30}C_{30})^2$

$$S = 30 \cdot ({}^{30}C_0)^2 + 29 \cdot ({}^{30}C_1)^2 + 28 \cdot ({}^{30}C_2)^2 + \dots + 0 \cdot ({}^{30}C_0)^2$$

$$2S = 30 \cdot ({}^{30}C_0^2 + {}^{30}C_1^2 + \dots + {}^{30}C_{30}^2)$$

$$S = 15 \cdot {}^{60}C_{30} = 15 \cdot \frac{60!}{(30!)^2}$$

$$\frac{15 \cdot 10!}{(30!)^2} = \frac{\alpha \cdot 60!}{(30!)^2}$$

$$\Rightarrow \alpha = 15$$

275. If a_r is the coefficient of x^{10-r} in the Binomial expansion of $(1+x)^{10}$, then $\sum_{r=1}^{10} r^3 \left(\frac{a_r}{a_{r-1}} \right)^2$ is equal to :

यदि $(1+x)^{10}$ के द्विपद प्रसार में x^{10-r} का गुणांक a_r है, तो $\sum_{r=1}^{10} r^3 \left(\frac{a_r}{a_{r-1}} \right)^2$ बराबर है :

- (A) 4895 (B) 5445 (C) 3025 (D) 1210

Ans. D

Sol. Coeff. of x^{10-r} in $(1+x)^{10}$

$$a_r = {}^{10}C_r$$

$$\frac{a_r}{a_{r-1}} = \frac{{}^{10}C_r}{{}^{10}C_{r-1}} = \frac{10-r+1}{r} = \frac{11-r}{r}$$

$$\sum_{r=0}^{10} r^3 \left(\frac{11-r}{r} \right)^2 = \sum_{r=0}^{10} r(11-r)^2$$

$$= \sum_{r=0}^{10} (r^3 - 22r^2 + 121r)$$

$$= \sum_{r=0}^{10} r^3 - 22 \sum_{r=0}^{10} r^2 + 121 \sum_{r=0}^{10} r = \left(\frac{10(10+1)}{2} \right)^2 - 22 \times \frac{10(10+1)(20+1)}{6} + 121 \times \frac{10 \cdot 11}{2}$$

$$= 55^2 - 11 \times 11 \times 70 + 5 \times 11^3 = 11^2 [25 - 70 + 55] = 11^2 \times 10 = 1210$$

276. Let K be the sum of the coefficients of the odd powers of x in the expansion of $(1+x)^{99}$. Let a be the middle term

in the expansion of $\left(2 + \frac{1}{\sqrt{2}} \right)^{200}$. If $\frac{{}^{200}C_{99} K}{a} = \frac{2^\ell m}{n}$, where m and n are odd numbers, then the ordered pair

(ℓ, n) is equal to

माना $(1+x)^{99}$ के प्रसार में x की विषम घातों के गुणांकों का योग K है। माना $\left(2 + \frac{1}{\sqrt{2}} \right)^{200}$ के प्रसार में मध्य पद a है। यदि

$\frac{{}^{200}C_{99} K}{a} = \frac{2^\ell m}{n}$ है, जहाँ m तथा n विषम संख्याएँ हैं, तो क्रमित युग्म (ℓ, n) बराबर है।

- (A) (50, 51) (B) (51, 101) (C) (50, 101) (D) (51, 99)

Ans. C

Sol. Sum of odd coeff. of $x = \frac{2^{99} - 0}{2} = k$

$$k = 2^{98}$$

$$\text{Middle term } \alpha = {}^{200}C_{100} \frac{(2)^{100}}{2^{50}}$$

$$= {}^{200}C_{100} 2^{50}$$

$$\text{Now } \frac{{}^{200}C_{99} \times k}{\alpha} = \frac{200!}{100!99!} \times \frac{2^{98}}{{}^{200}C_{100} \times 2^{50}}$$

$$= \frac{100}{101} \times 2^{48}$$

$$\frac{25}{101} \times 2^{50}$$

277. The coefficient of x^{301} in $(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \dots + x^{500}$ is :

$(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \dots + x^{500}$ में x^{301} का गुणांक है :

- (A) ${}^{501}C_{302}$ (B) ${}^{500}C_{300}$ (C) ${}^{500}C_{301}$ (D) ${}^{501}C_{200}$

Ans. D

Sol. $(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \dots + x^{500}$

$$= (1+x)^{500} \left\{ \frac{1 - \left(\frac{x}{1+x} \right)^{501}}{1 - \frac{x}{1+x}} \right\}$$

$$= (1+x)^{500} \frac{\left((1+x)^{501} - x^{501} \right)}{(1+x)^{501}} \cdot (1+x)$$

$$= (1+x)^{501} - x^{501}$$

Coefficient of x^{301} in $(1+x)^{501} - x^{501}$ is given by ${}^{501}C_{301} = {}^{501}C_{200}$

278. Let $x = (8\sqrt{3} + 13)^{13}$ and $y = (7\sqrt{2} + 9)^9$. If $[t]$ denotes the greatest integer $\leq t$, then

- (A) $[x]$ is even but $[y]$ is odd (B) $[x] + [y]$ is even
(C) $[x]$ and $[y]$ are both odd (D) $[x]$ is odd but $[y]$ is even

माना $x = (8\sqrt{3} + 13)^{13}$ और $y = (7\sqrt{2} + 9)^9$ है। यदि $[t]$ महत्तम पूर्णांक $\leq t$ है, तब

(A) $[x]$ सम है परन्तु $[y]$ विषम है

(B) $[x] + [y]$ सम है

(C) $[x]$ एवं $[y]$ दोनों विषम है

(D) $[x]$ विषम है परन्तु $[y]$ सम है

Ans. B

Sol. $x = (8\sqrt{3} + 13)^{13} = {}^{13}C_0 (8\sqrt{3})^{13} + {}^{13}C_1 (8\sqrt{3})^{12} + \dots$

$$x' = (8\sqrt{3} - 13)^{13} = {}^{13}C_0 (8\sqrt{3})^{13} - {}^{13}C_1 (8\sqrt{3})^{12} (13)^1 + \dots$$

$$x - x' = 2 \left[{}^{13}C_1 (8\sqrt{3})^{12} (13)^1 + {}^{13}C_3 (8\sqrt{3})^{10} \cdot (13)^3 + \dots \right]$$

therefore, $x - x'$ is even integer, hence $[x]$ is even

Now, $y = (7\sqrt{2} + 9)^9 = {}^9C_0 (7\sqrt{2})^9 + {}^9C_1 (7\sqrt{2})^8 (9)^1 + {}^9C_2 (7\sqrt{2})^7 (9)^2 + \dots$

$$y' = (7\sqrt{2} - 9)^9 = {}^9C_0 (7\sqrt{2})^9 - {}^9C_1 (7\sqrt{2})^8 (9)^1 + {}^9C_2 (7\sqrt{2})^7 (9)^2 + \dots$$

$$y - y' = 2 \left[{}^9C_1 (7\sqrt{2})^8 (9)^1 + {}^9C_3 (7\sqrt{2})^6 (9)^3 + \dots \right]$$

$y - y'$ = Even integer, hence $[y]$ is even

279. The value of $\frac{1}{1!50!} + \frac{1}{3!48!} + \frac{1}{5!46!} + \dots + \frac{1}{49!2!} + \frac{1}{51!1!}$ is :

$$\frac{1}{1!50!} + \frac{1}{3!48!} + \frac{1}{5!46!} + \dots + \frac{1}{49!2!} + \frac{1}{51!1!} \text{ का मान है :}$$

(A) $\frac{2^{51}}{51!}$

(B) $\frac{2^{50}}{51!}$

(C) $\frac{2^{51}}{50!}$

(D) $\frac{2^{50}}{50!}$

Ans. B

Sol. $\sum_{r=1}^{26} \frac{1}{(2r-1)!(51-(2r-1))!} = \sum_{r=1}^{26} {}^{51}C_{(2r-1)} \frac{1}{51!}$

$$= \frac{1}{51!} \{ {}^{51}C_1 + {}^{51}C_3 + \dots + {}^{51}C_{51} \} = \frac{1}{51!} (2^{50})$$

280. The sum $\sum_{n=1}^{\infty} \frac{2n^2 + 3n + 4}{(2n)!}$ is equal to :

योगफल $\sum_{n=1}^{\infty} \frac{2n^2 + 3n + 4}{(2n)!}$ बराबर है :

(A) $\frac{13e}{4} + \frac{5}{4e}$

(B) $\frac{13e}{4} + \frac{5}{4e} - 4$

(C) $\frac{11e}{2} + \frac{7}{2e}$

(D) $\frac{11e}{2} + \frac{7}{2e} - 4$

Ans. B

Sol. $\sum_{n=1}^{\infty} \frac{2n^2 + 3n + 4}{2n!}$

$$= \frac{1}{2} \sum_{n=1}^{\infty} \frac{2n(2n-1) + 8n + 8}{2n!}$$

$$= \frac{1}{2} \sum_{n=1}^{\infty} \left\{ \frac{1}{(2n-2)!} + \frac{4}{(2n-1)!} + \frac{8}{2n!} \right\}$$

$$= \frac{1}{2} \left\{ \left(\frac{e + e^{-1}}{2} \right) + 4 \left(\frac{e - e^{-1}}{2} \right) + 8 \left(\frac{e + e^{-1}}{2} - 1 \right) \right\}$$

$$= \frac{1}{4} (13e + 5e^{-1} - 16)$$

$$= \frac{13}{4}e + \frac{5}{4e} - 4$$